
Exercise 6

1 Robust Invariant Set

1. Consider the following scalar autonomous discrete-time system

$$x^+ = 0.5x + w$$

under the constraint $x \in \mathbb{X} = [-10, 10]$ and subject to a bounded disturbance $w \in \mathbb{W} = [-1, 1]$. Compute the maximum robust invariant set.

2. Suppose that $\mathbb{Y}$ is a robust control invariant set for the system $x^+ = Ax + Bu + w$ subject to the state and input constraints $\mathbb{X}$ and $\mathbb{U}$ and the noise $w \in \mathbb{W}$, where 0 is in $\mathbb{W}$, $\mathbb{X}$ and $\mathbb{U}$. For which of the following scenarios is $\mathbb{Y}$ still a robust control invariant set for the resulting system?
 - (a) Disturbance set changes to $\alpha\mathbb{W}$ for some $\alpha \in \mathbb{R}_+$.
 - (b) Input constraint set changes to $\alpha\mathbb{U}$ for some $\alpha \in \mathbb{R}_+$.

2 Robust linear MPC for nonlinear system

Consider the following nonlinear discrete-time dynamic system

$$x^+ = rx(1 - x^2) + u = f(x, u)$$

subject to the state and input constraints $x \in \mathbb{X} = [-0.5, 0.5]$ and $u \in \mathbb{U} = [-1, 1]$. Treat r symbolically as a constant $r \in (0, 0.5)$. Design a robust linear MPC controller for this nonlinear system to regulate it to the origin with the following steps:

1. Find values a, b by linearizing the system around the origin and the smallest $\bar{w}$ such that the evolution of the following uncertain linear dynamic system contains that of the nonlinear system above for all $x \in \mathbb{X}, u \in \mathbb{U}$.

$$x^+ = ax + bu + w \quad w \in \mathbb{W} = \bar{w} \cdot [-1, 1]$$

2. Compute the maximum robust invariant set $\mathbb{X}_f$ for the linearized system within the constraint set $\mathbb{X}$ for the control law $u = 0$. Use the value $\bar{w} = 0.1$ for this question.
3. Design a set $C(x)$ such that any controller $u(x) \in C(x)$ will ensure robust constraint satisfaction in two stages of open-loop control of the linearized system. Use the value $\bar{w} = 0.1$ for this question.
Hint: the set $C(x)$ has the shape of

$$C(x) := \left\{ u_0 \mid \exists u_1 \text{ s.t. } -h \leq G \begin{bmatrix} u_0 \\ u_1 \\ x \end{bmatrix} \leq h \right\}$$

where x is the current state, u_0 is the next input to apply, and u_1 is the planned input in the next stage. The problem is converted to finding a proper matrix-vector pair of (G, h) .